

Unit 4, Lesson 11: Applying Multiple Laws of Exponents – Part 2

Benchmark

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Learning Target

- ☐ I can write an equivalent algebraic expression by applying multiple laws of exponents.

4.11.1 Warm-Up: Reading Numbers

1. In the U.S., the number 42,067,513,708 is read as, "42 billion, 67 million, 513 thousand, 708."

In many Latin American countries and in the United Kingdom, the same number is read as, "42 thousand million, 67 million, 513 thousand, 708."

- Explain or show how 42 billion and 42 thousand million describe the same quantity.
- Express the quantity 42 thousand million using scientific notation.

4.11.2 Activity: Riddle Me This

1. Work with your partner to generate equivalent expressions for each expression given below by applying the laws of exponents. Your results will be used to complete the riddle in question 2.

E $a^2 \cdot (a^4)^{-2}$	N $\frac{a^3}{a \cdot a^4}$
O $\frac{(a^3)^4}{a^2}$	Y $a^{-3} \cdot a \cdot (a^2)^3$
P $\left(\frac{a^{-2}}{a^4}\right)^3 \cdot a$	N $\frac{(a^{-2})^{-4}}{a^3 \cdot a^3}$
E $\frac{a^4}{(2a^4)^3}$	O $(4a)(2a^{-3})^{-4}$
T $-2a^{-3} - (-2a^3)^2$	X $\frac{(5a^7)^3}{a} \cdot \left(\frac{a^6}{2}\right)^{-2}$
N $\frac{-9a^4b^7}{2a^3b^4}$	T $(a^{-4}b \cdot a^2b^{-1})^2$
N $(10a^3b^{-4})^2 \cdot (10a)^{-1}$	P $\frac{(2a^3b^{-4})^4}{(4a^4b^{-3})^3}$

2. Complete the answers to the riddle by writing the letters in the blanks from the expressions in question 1 that are equivalent to each expression.

In the expression x^3 , what do you call 3?

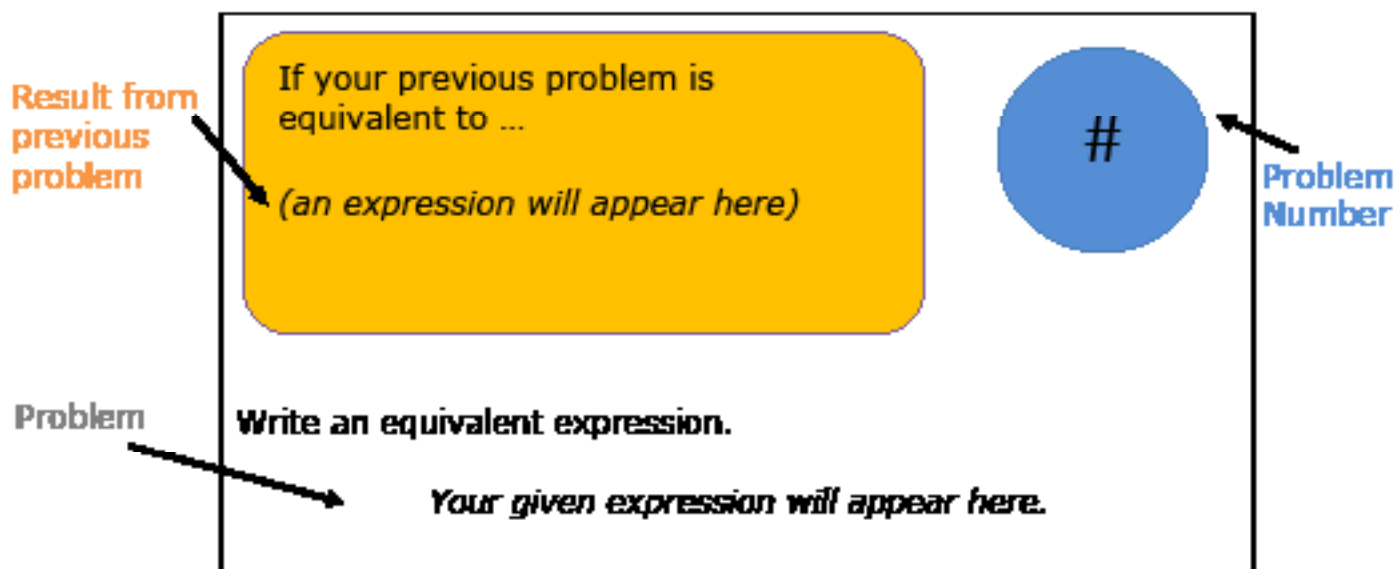
An _____ - _____
 $100a$ $\frac{1}{a^{37}}$ $\frac{a^{38}}{4}$ $\frac{1}{a^3}$ $\frac{1}{2a^6}$ $\frac{-2b}{a^3}$ $-8a$

In the expression y^2 , what do you call 2?

A _____ - _____
 a^4 a^3b^{11} a^6 $\frac{10a^3}{b^9}$ $\frac{1}{a^4}$ 1 $\frac{1}{a^4}$

4.11.3 Activity: Scavenger Hunt

Work with your partner or group to complete the scavenger hunt. Each card in the scavenger hunt appears as shown.



Write the number of the problem you start with and the given expression in the first row of the table. Then, work with your group or partner to generate an equivalent expression using the fewest number of factors.

Look for the card with an expression equivalent to your problem to find your next problem to work on.

Repeat this process until your group has completed all the scavenger hunt problems and have ended up back at your original problem.

[illegible]

4.11.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning target in the table.

 I need help.

 I got it.

 I could help someone.

Apply one law of exponents to generate an equivalent algebraic expression	  
Apply multiple laws of exponents to generate an equivalent algebraic expression	  
Determine if two algebraic expressions are equivalent	  

